

Cross-Regime Measurement Failure in the KT3b Environmental Test

Morten Magnusson

Independent Researcher, Sola, Norway

ORCID: 0009-0002-4860-5095

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Abstract. *The KT3b environmental test of Energy-Flow Cosmology returned a null result after four iterations of increasing methodological rigour. We show that the observed monotonic signal collapse across these iterations is consistent with a cross-regime measurement inconsistency, and cannot be uniquely explained by statistical limitations alone. The theoretical prediction (mode activation D_{eff}) lives in the non-linear galactic regime L2, the environmental driver (ρ_{env}) lives in the linear cosmological regime L1, the observable ($\mathcal{R}$ at $g \approx a_0$) sits on the L1/L2 boundary, and the coupling between regimes (α) is not defined as an operator mapping L1 density fields to L2 dynamical response. This constitutes a violation of the Regime-Consistent Measurement Principle (RCMP). We identify the structural failure, classify three valid test architectures, and argue that the diagnosis is a stronger contribution than the null result itself.*

1 The KT3b Measurement Chain

KT3b attempted to detect the environmental coupling parameter α in

$$\rho_{\text{eff}} = \rho_{\text{gal}}(r) + \alpha \rho_{\text{env}} \quad (1)$$

by measuring the MDAR residual $\mathcal{R} = g_{\text{obs}}/g_{\text{bar}}$ at $g_{\text{bar}} \approx a_0$ and correlating it with tidal environment Θ_1 (or group membership N_{mem}).

Four runs were executed with increasing control:

Table 1: KT3b run history. Signal collapses as methodology improves.

Run	Env. proxy	N	Slope	Diagnosis
1	$\log(L/D^2)$	~ 60	-0.21	Mass artefact
2	Karachentsev Θ_1	17	$+0.01$	N too small
3	Tully N_{mem} (raw)	70	-0.07	Mass bias
4	Tully (mass-matched pairs)	23	≈ 0	Null

The pattern is monotonic: as confounders are removed, the signal vanishes. The standard interpretation is “effect too small for SPARC.” We argue for a deeper diagnosis.

2 The Regime Mismatch

Every component in the KT3b measurement chain occupies a different regime in the EFC hierarchy:

Table 2: Regime assignment of KT3b components.

Component	Regime	Why
$D_{\text{eff}}(\rho)$ (theory)	L2	Non-linear galactic dynamics; grid-mode activation at $g \lesssim a_0$
ρ_{env} (driver)	L1	Large-scale structure; Mpc-scale density field; linear perturbation theory
$\mathcal{R}$ at a_0 (observable)	L1/L2 boundary	The transition point where the regime changes; maximally unstable
Rotation curve (data)	L2	kpc-scale circular velocities
Υ_* (mass model)	L1 assumption	Stellar population synthesis assumes universal IMF and standard gravity

The test attempts to measure an L2 effect using an L1 driver, observed at the L1/L2 boundary. This is a *cross-regime measurement*: the information must propagate from L1 through the regime boundary into L2 and back out through $\mathcal{R}$.

2.1 Why the boundary is the worst place to measure

The L1/L2 transition occurs where $g \approx a_0$. This is precisely where:

- The stiffness response $R(k)$ transitions from $R \gg 1$ (L1, suppressed growth) to $R \ll 1$ (L2, enhanced gravity) [1].
- The entropy production function $\Gamma(\rho)$ crosses its inflection point [2].
- The effective Poisson coupling $\mu(k)$ swings from < 1 to > 1 [1].
- Observational scatter in the RAR is maximal [3].

Measuring at a_0 is analogous to measuring a phase transition *at* the critical temperature: fluctuations diverge and signal-to-noise collapses. More precisely, at the L1/L2 boundary the fluctuations in the response function $\mu(k)$ dominate over the mean signal from any cross-regime coupling, implying that $\mathcal{R}(a_0)$ has intrinsically low Fisher information with respect to cross-regime parameters such as α .

2.2 The undefined coupling operator

The parameter α in Eq. (1) is not a standard model parameter—it is a *cross-regime coupling operator* that maps L1 density fields to L2 dynamical response. In the current formulation:

- α is not defined as an operator in function space: without a scale- and mode-dependent kernel $K(k, \rho_{\text{env}})$, the coupling between L1 and L2 is underdetermined.
- It has no scale dependence: the same α applies regardless of the spatial scale separating L1 and L2.
- It does not specify the *mechanism* by which Mpc-scale density affects kpc-scale mode activation.

Without a derived coupling operator, the test is structurally underdetermined: any measured slope b could be attributed to α , to an L1/L2 projection artefact, or to residual

mass–environment correlation. In the absence of a kernel $K(k, z)$, the mapping $\rho_{\text{env}} \mapsto D_{\text{eff}}$ is not unique, and α cannot be interpreted as a physical parameter but only as an effective fit coefficient absorbing all cross-regime effects indiscriminately.

3 RCMP Violation

The Regime-Consistent Measurement Principle [4] requires that:

The theoretical prediction, the observable, and the measurement instrument must all be defined within, or have explicit and derived mappings between, the same regime(s).

RCMP is not an auxiliary assumption but a consistency requirement: without a regime-consistent mapping, the observable is not a well-defined functional of the theoretical degrees of freedom, and the measurement carries no unambiguous information about the parameters of interest.

KT3b violates this in three ways:

- (V1) **Driver–response mismatch.** ρ_{env} (L1) drives D_{eff} (L2) through an underived α .
- (V2) **Boundary observation.** $\mathcal{R}$ at a_0 sits on the L1/L2 boundary where both regimes contribute and neither dominates.
- (V3) **Implicit regime mixing in mass model.** $\Upsilon_* = 0.5 M_{\odot}/L_{\odot}$ assumes L1-regime stellar physics applied to L2-regime gravitational dynamics.

Any one of these violations is sufficient to degrade the measurement. Together, they explain the systematic collapse of the signal across all four runs.

4 Three Valid Test Architectures

A regime-consistent test must keep all components within one regime, or provide explicit derived mappings between regimes.

4.1 Architecture A: Pure L2 (galactic interior)

Prediction: D_{eff} modifies the *shape* of the RAR in the deep-MOND regime ($g \ll a_0$), not just its value at a_0 . If environmental density modulates D_{eff} , the RAR *slope* at $g/a_0 \sim 0.01\text{--}0.1$ should correlate with environment.

Observable: $d \log(g_{\text{obs}}/g_{\text{bar}})/d \log g_{\text{bar}}$ evaluated at $g_{\text{bar}} \ll a_0$ (deep L2 regime).

Driver: Same Θ_1 or N_{mem} , but now the observable is safely inside L2.

Why it works: The measurement point is far from the regime boundary. Unlike $\mathcal{R}(a_0)$, the deep-MOND slope probes a regime where the response function has saturated ($\mu \gg 1$), reducing sensitivity to boundary fluctuations and isolating the intrinsic L2 dynamics. The RAR slope in deep MOND is a stable L2 quantity with lower intrinsic scatter than $\mathcal{R}$ at a_0 , and is directly sensitive to the functional form of D_{eff} rather than its value at a single point.

4.2 Architecture B: Pure L1 (cosmological growth)

Prediction: D_{eff} enters the cosmological background through $\Gamma(\rho)$, which controls the growth rate $f\sigma_8(z)$. Environmental density at L1 scales directly modulates Γ without cross-regime coupling.

Observable: $f\sigma_8(z)$ from redshift-space distortions, or void–cluster growth rate comparison.

Why it works: Everything is in L1. The MCMC inference pipeline already constrains α_{L1} (currently at -0.14 ± 0.21).

4.3 Architecture C: Derived L1→L2 mapping

Requirement: Derive α from the EFC action by computing how the background S -field at L1 scales enters the boundary conditions of the L2 Poisson equation.

Observable: Same as KT3b, but with a *predicted* $\alpha(k, z)$ rather than a free parameter.

Why it is hard: This requires solving the coupled $(g_{\mu\nu}, S)$ system across scales, which is an open problem in EFC.

Table 3: Comparison of test architectures.

	A (pure L2)	B (pure L1)	C (derived)
RCMP-consistent	Yes	Yes	Yes
Data available now	Yes (SPARC)	Partial ($f\sigma_8$)	No
Free parameters	0	1 (α_{L1})	0
Discriminates EFC/MOND	Yes (shape)	Yes (growth)	Yes (amplitude)

5 Conclusion

The KT3b null result is not a failure of the physical hypothesis (D_{eff} environmental coupling) or of the dataset (SPARC). The observed monotonic signal collapse across four runs of increasing rigour is consistent with a cross-regime measurement design that violates RCMP:

- (1) The driver (ρ_{env}) and the response (D_{eff}) live in different regimes.
- (2) The observable ($\mathcal{R}$ at a_0) sits on the regime boundary where signal-to-noise is structurally minimised.
- (3) The coupling (α) is not defined as an operator mapping L1 fields to L2 response, leaving the cross-regime information channel underdetermined.

The observable $\mathcal{R}(a_0)$ is not a regime-invariant quantity, and therefore cannot carry unambiguous information about cross-regime couplings. This is the fundamental reason the signal collapses as methodology improves: better control removes projection artefacts, but the remaining physical information is not accessible at the regime boundary.

The diagnosis is more valuable than the null. It identifies a structural constraint on *any* test that attempts to connect large-scale environment to galactic dynamics in a regime-structured theory. The same constraint applies to MOND’s External Field Effect, Verlinde’s emergent gravity, and any framework where the transition scale a_0 marks a regime boundary.

Architecture A (pure L2, RAR slope at $10^{-2} \lesssim g_{\text{bar}}/a_0 \lesssim 10^{-1}$) is the immediate next step: it uses existing SPARC data, requires no free parameters, and keeps the entire measurement within a single regime.

References

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